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(54) **METHODS AND APPARATUS FOR PERFORMING SPINE SURGERY**

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(58) **Field of Classification Search**

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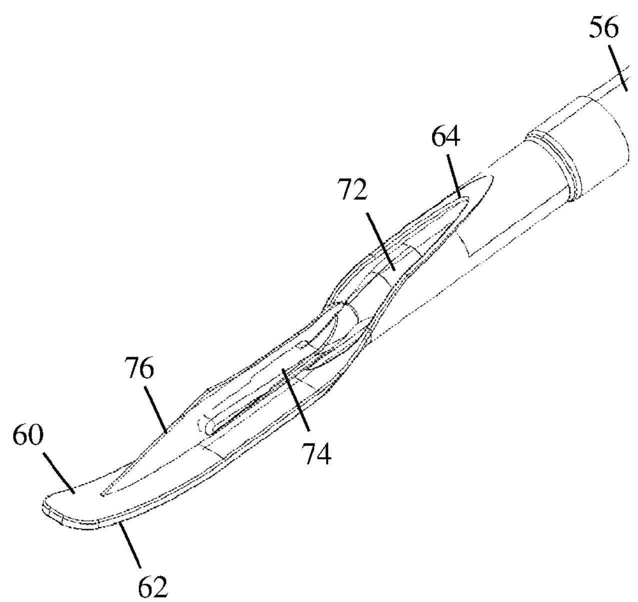
(57) **ABSTRACT**

Systems and methods are described for correcting sagittal imbalance in a spine including instruments for performing the controlled release of the anterior longitudinal ligament through a lateral access corridor and hyper-lordotic lateral implants.

17 Claims, 7 Drawing Sheets



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continuation of application No. 15/702,685, filed on Sep. 12, 2017, now Pat. No. 10,426,627, which is a continuation of application No. 14/918,137, filed on Oct. 20, 2015, now Pat. No. 9,757,246, which is a continuation of application No. 14/578,215, filed on Dec. 19, 2014, now Pat. No. 9,192,482, which is a continuation of application No. 13/653,335, filed on Oct. 16, 2012, now Pat. No. 8,920,500, which is a continuation of application No. 12/799,021, filed on Apr. 16, 2010, now Pat. No. 8,287,597.

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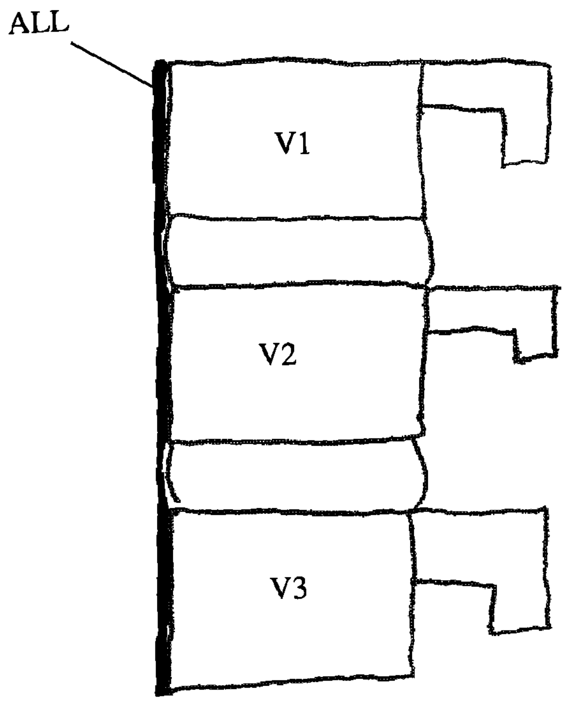

FIG. 1

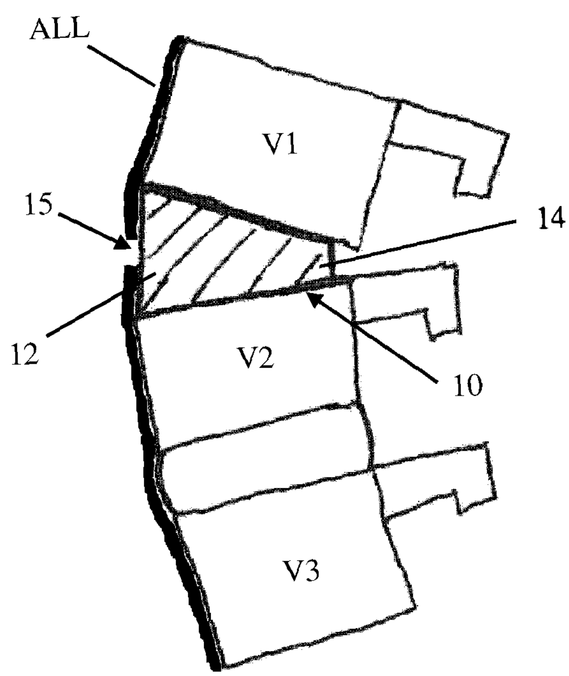

FIG. 2

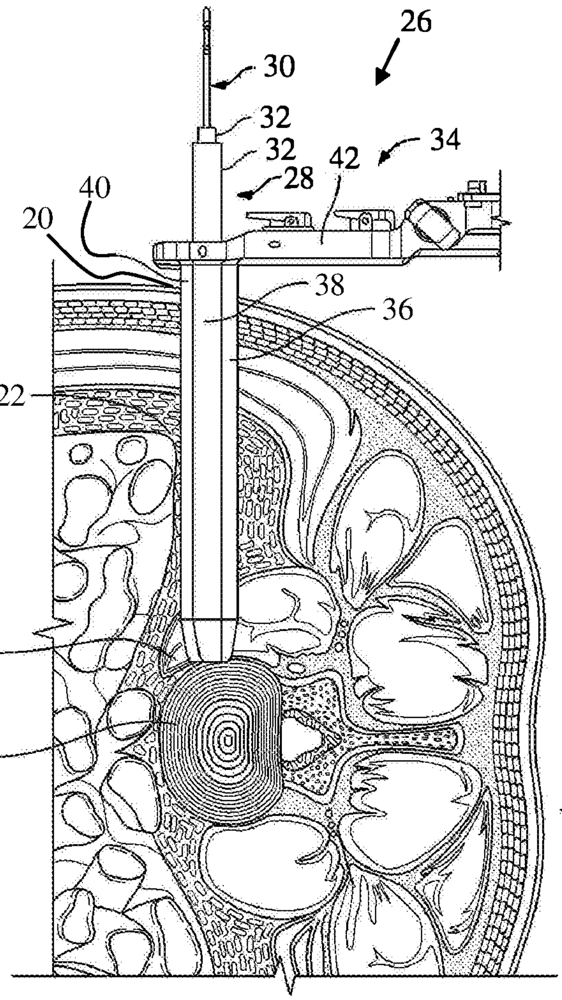

FIG. 3

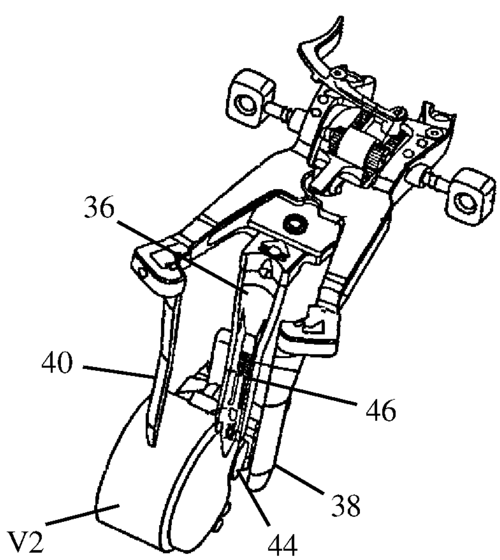

FIG. 4

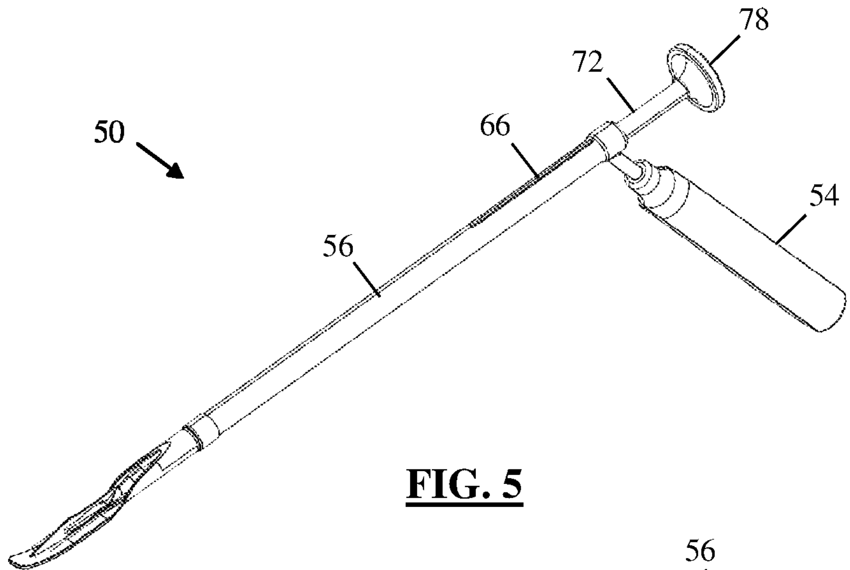

FIG. 5

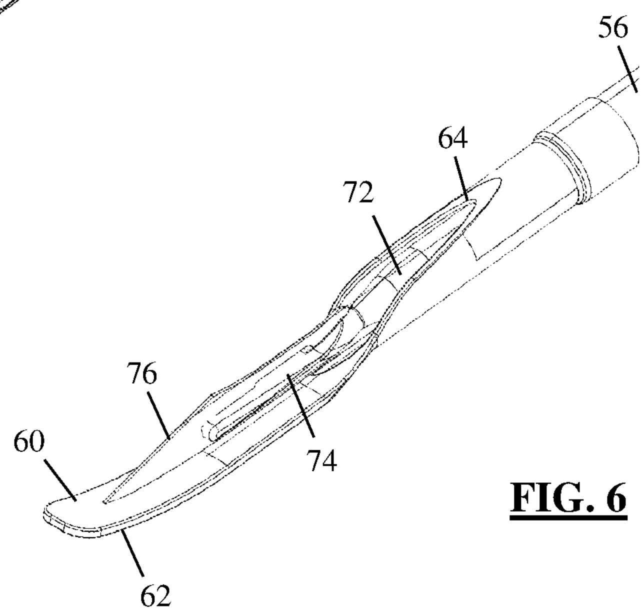

FIG. 6

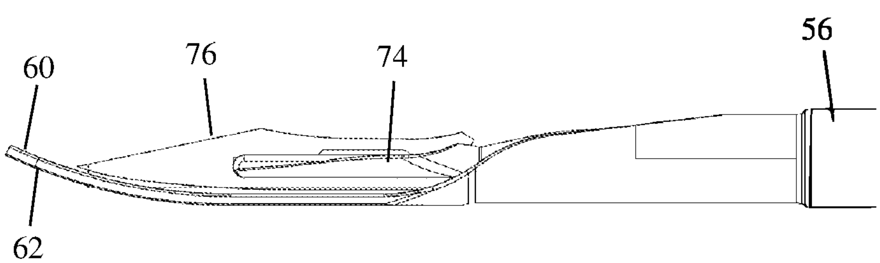

FIG. 7

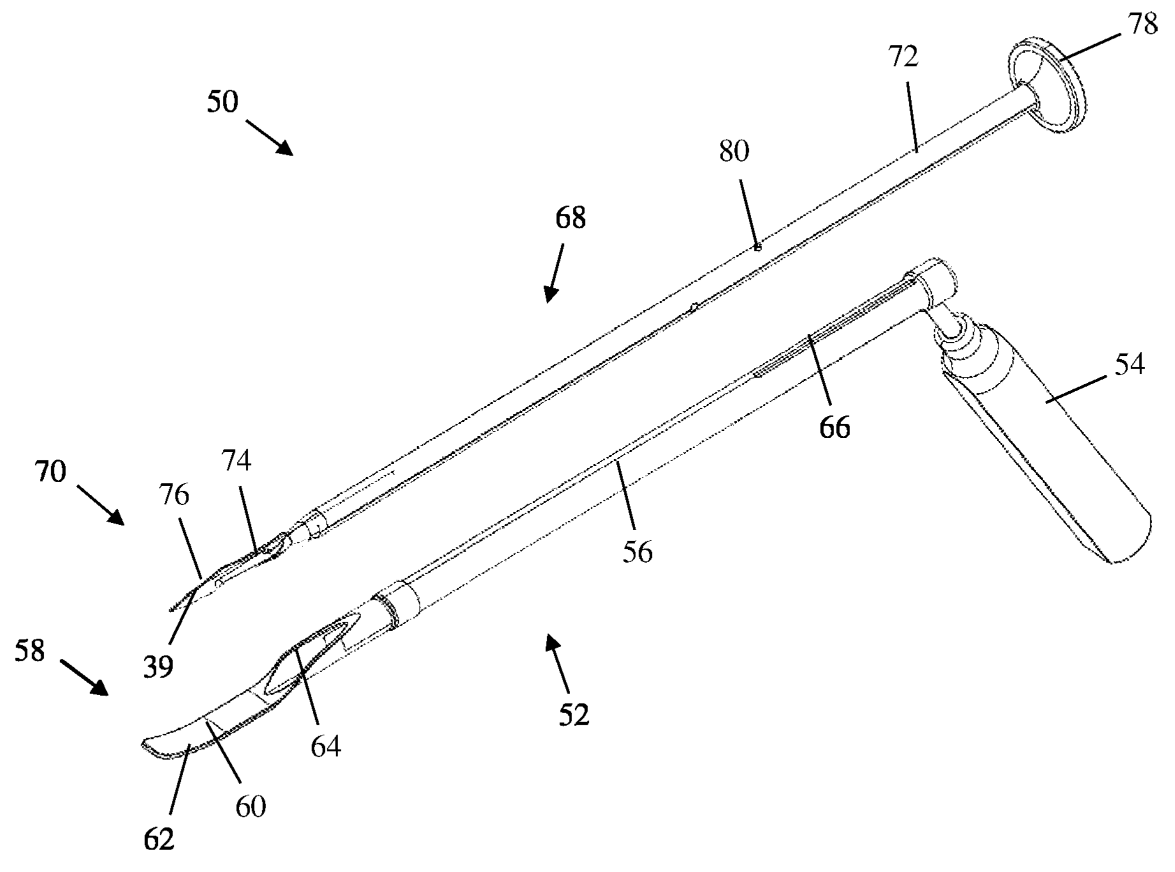

FIG. 8

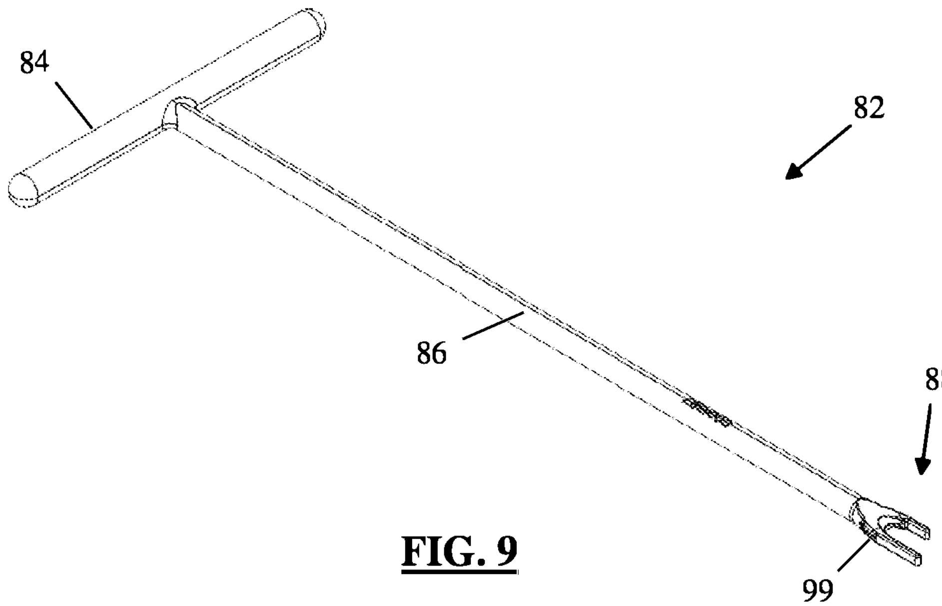

FIG. 9

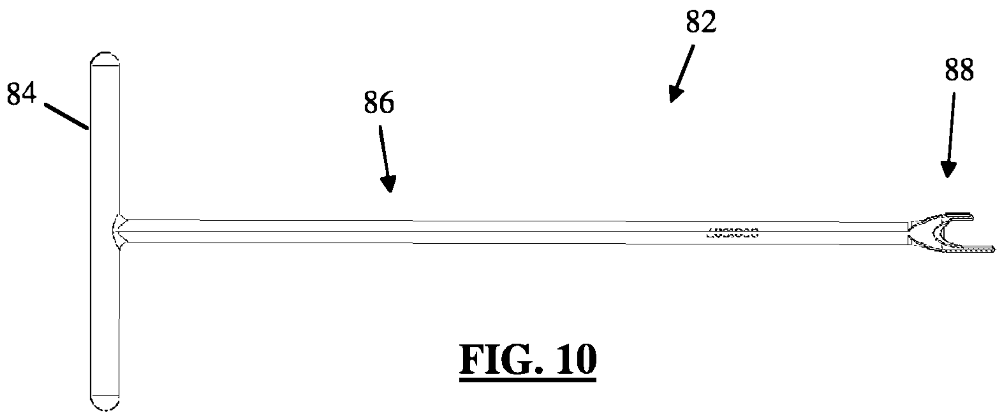

FIG. 10

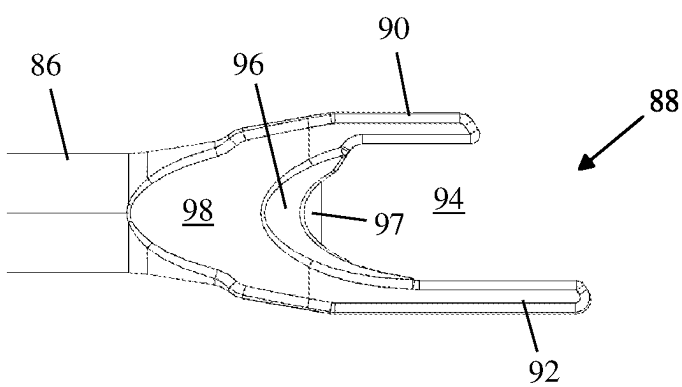

FIG. 11

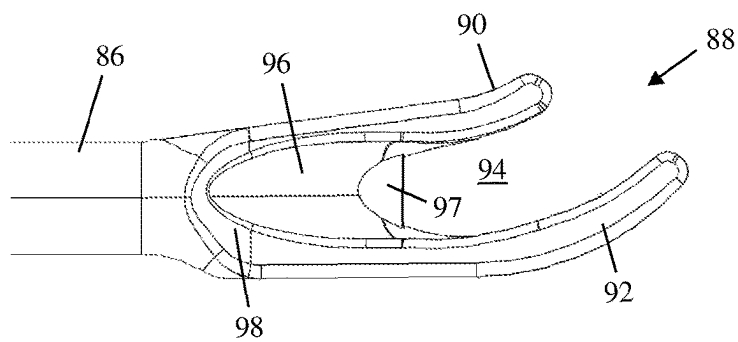

FIG. 12

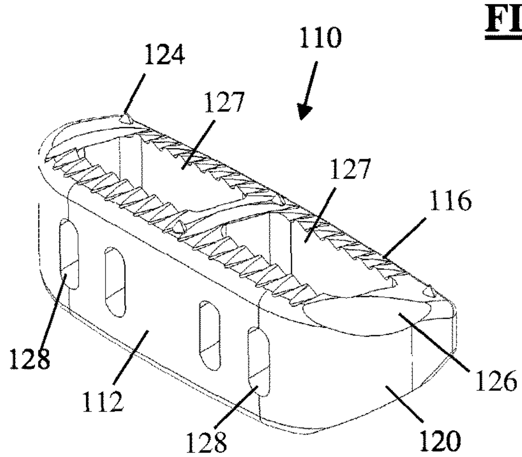

FIG. 13

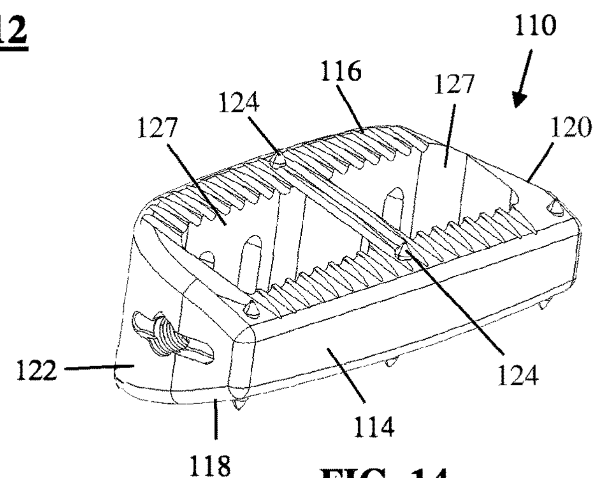

FIG. 14

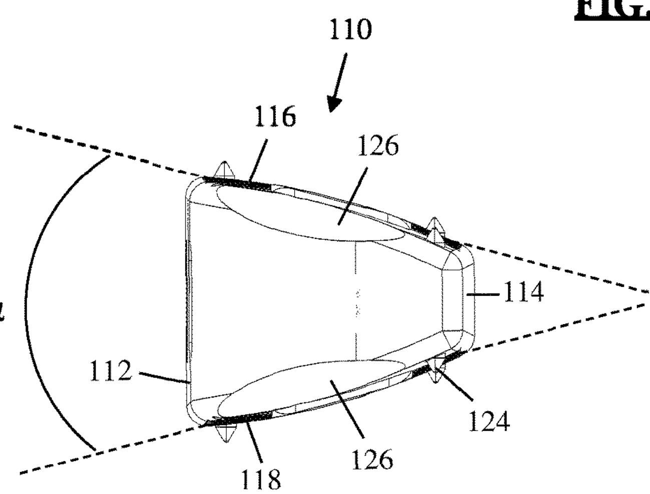

FIG. 15

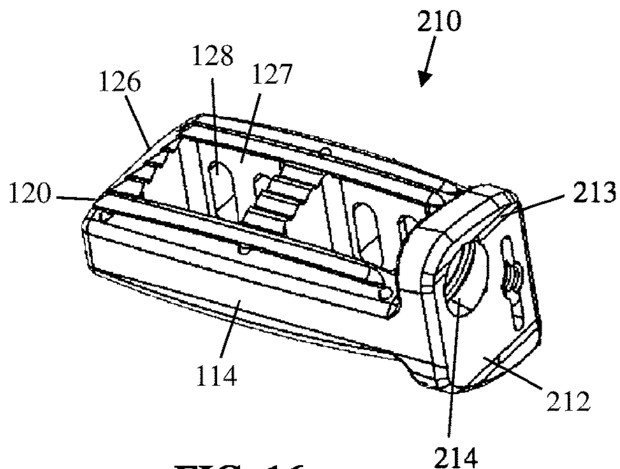

FIG. 16

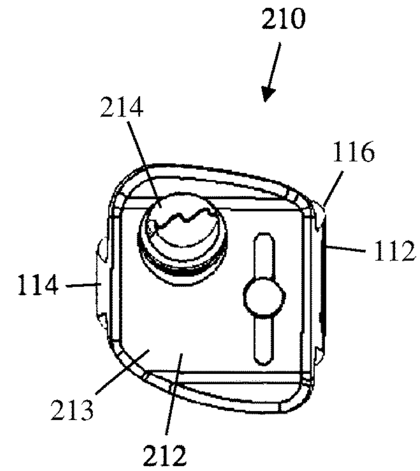

FIG. 17

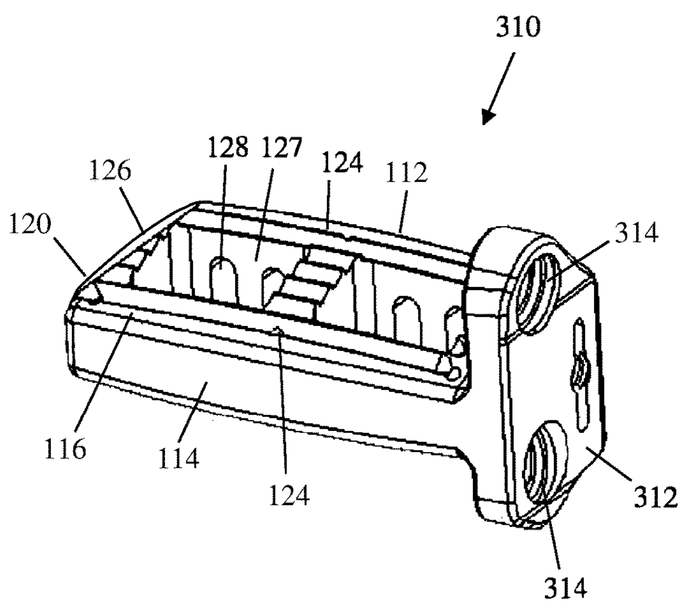

FIG. 18

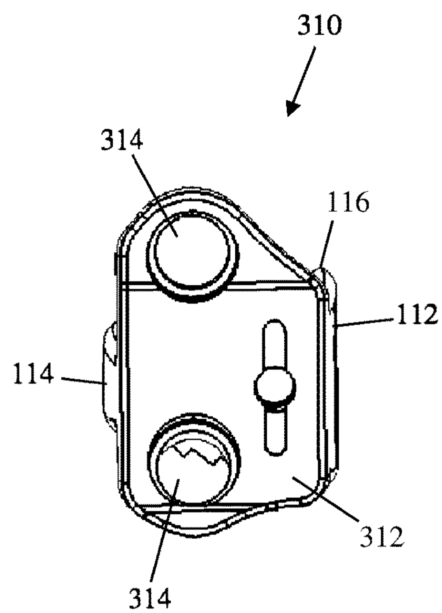

FIG. 19

METHODS AND APPARATUS FOR PERFORMING SPINE SURGERY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of co-pending U.S. patent application Ser. No. 16/545,821 filed Aug. 20, 2019, which is a continuation of U.S. patent application Ser. No. 15/702,685 filed Sep. 12, 2017 (now U.S. Pat. No. 10,426,627), which is a continuation of U.S. patent application Ser. No. 14/918,137 filed Oct. 20, 2015 (now U.S. Pat. No. 9,757,246), which is a continuation of U.S. patent application Ser. No. 14/578,215 filed Dec. 19, 2014 (now U.S. Pat. No. 9,192,482), which is a continuation of U.S. patent application Ser. No. 13/653,335 filed Oct. 16, 2012 (now U.S. Pat. No. 8,920,500), which is a continuation of U.S. patent application Ser. No. 12/799,021 filed Apr. 16, 2010 (now U.S. Pat. No. 8,287,597), which claims the benefit of priority from U.S. Provisional Patent Application Ser. No. 61/212,921 filed Apr. 16, 2009, and U.S. Provisional Patent Application Ser. No. 61/319,823 filed Mar. 31, 2010, the entire contents of which is hereby expressly incorporated by reference into this disclosure as if set forth in its entirety herein.

FIELD

The present invention relates to implants and methods for adjusting sagittal imbalance of a spine.

BACKGROUND

A human spine has three main regions—the cervical, thoracic, and lumbar regions. In a normal spine, the cervical and lumbar regions have a lordotic (backward) curvature, while the thoracic region has a kyphotic (forward) curvature. Such a disposition of the curvatures gives a normal spine an S-shape. Sagittal imbalance is a condition in which the normal alignment of the spine is disrupted in the sagittal plane causing a deformation of the spinal curvature. One example of such a deformity is "flat-back" syndrome, wherein the lumbar region of the spine is generally linear rather than curved. A more extreme example has the lumbar region of the spine exhibiting a kyphotic curvature such that the spine has an overall C-shape, rather than an S-shape. Sagittal imbalance can be a problem in that it is biomechanically disadvantageous and generally results in discomfort, pain, and an awkward appearance in that the patient tends to be bent forward excessively.

Various treatments for sagittal imbalance are known in the art. These treatments generally involve removing at least some bone from a vertebra (osteotomy) and sometimes removal of the entire vertebra (vertebrectomy), in order to reduce the posterior height of the spine in the affected region and recreate lordotic curve. Such procedures are traditionally performed via an open, posterior approach involving a large incision (often to expose multiple spinal levels at the same time) and require stripping of the muscle tissue away from the bone. These procedures can have the disadvantages of a large amount of blood loss, high risk, and a long and painful recovery for the patient. Furthermore, depending upon the patient, multiple procedures, involving both anterior and posterior approaches to the spine, may be required.

The present invention is directed at overcoming, or at least improving upon, the disadvantages of the prior art.

BRIEF DESCRIPTION OF THE DRAWINGS

Many advantages of the present invention will be apparent to those skilled in the art with a reading of this specification in conjunction with the attached drawings, wherein like reference numerals are applied to like elements and wherein:

FIG. **1** is a lateral view representing a portion of a sagitally imbalanced lumbar spine lacking the normal lordotic curvature;

FIG. **2** is a side view representing the lumbar spine of FIG. **1** after restoration of the lordotic curvature using a hyper-lordotic fusion implant, according to one example embodiment;

FIG. **3** is a top down view depicting the creation of a lateral access corridor formed with surgical access system through the side of the patient to the target disc space, according to one example embodiment;

FIG. **4** is a perspective view depicting a lateral access corridor formed with a retractor assembly through the side of the patient to the target disc space, according to one example embodiment;

FIG. **5** is a front perspective view of an Anterior Longitudinal Ligament (ALL) resector for safely releasing the ALL through a lateral access corridor, according to one example embodiment;

FIG. **6** is an enlarged from perspective view of the distal portion of the ALL resector of FIG. **5**;

FIG. **7** is an enlarged side view of the distal portion of the ALL resector of FIG. **5**;

FIG. **8** is an exploded front perspective view of the ALL resector of FIG. **5**;

FIG. **9** is a front perspective view of an ALL resector for safely releasing the ALL through a lateral access corridor, according to another example embodiment;

FIG. **10** is a side view of the ALL resector of FIG. **9**;

FIG. **11** is an enlarged side view of the distal end of the ALL resector of FIG. **9**;

FIG. **12** is an enlarged side view of the distal end of the ALL resector of FIG. **9** having curved finger portions;

FIGS. **13-15** are anterior-front perspective, posterior-rear perspective, and front views of a hyper-lordotic lateral implant according to one example embodiment;

FIGS. **16-17** are posterior-rear perspective and rear views of a hyper-lordotic lateral implant according to another example embodiment; and

FIGS. **18-19** are posterior-rear perspective and rear views of a hyper-lordotic lateral implant according to still another example embodiment.

DESCRIPTION OF A PREFERRED EMBODIMENT

Illustrative embodiments of the invention are described below. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure. The methods and devices described herein include a variety of inventive features and components that warrant patent protection, both individually and in combination.

With reference to FIGS. **1**-**2**, devices and methods of the present invention are utilized to correct sagittal imbalance, including lumbar kyphosis, by increasing the anterior height of the affected spinal area (as opposed to reducing the posterior height). FIG. **1** illustrates a portion of the lumbar spine lacking the standard lordotic curvature. To correct the sagittal imbalance, illustrated in FIG. **2**, a hyper-lordotic implant **10** is positioned into the disc space at the appropriate spinal level (e.g. between V1 and V2). An anterior sidewall **12** of hyper-lordotic implant **10** has a height significantly larger than an opposing posterior sidewall **14** such that when the implant is positioned within the disc space the anterior aspects of V1 and V2 are forced apart while the posterior aspects are not (or at least not to the same degree), thus imparting a lordotic curvature into the spine. To allow the anterior aspects of V1 and V2 to separate and receive the hyper-lordotic implant **10**, the Anterior Longitudinal Ligament (ALL) that runs along the anterior aspect of the spine may be released or cut (**15**). According to a preferred method, the implant **10** is implanted through a lateral access corridor formed through the side of the patient. Accessing the targeted spinal site through the lateral access corridor avoids a number of disadvantages associated with posterior access (e.g. cutting through back musculature and possible need to reduce or cut away part of the posterior bony structures like lamina, facets, and spinous process) and anterior access (e.g. use of an access surgeon to move various organs and blood vessels out of the way in order to reach the target site). Accordingly, by accessing the target site via a lateral access approach and correcting the sagittal imbalance without reducing the posterior height (i.e. no bone removal) the high blood loss and painful recovery associated previous methods may be avoided (or at least mitigated).

According to one example, the lateral access approach to the targeted spinal space may be performed according to the instruments and methods described in commonly owned U.S. Pat. No. 7,207,949 entitled "Surgical Access System and Related Methods," and/or U.S. patent application Ser. No. 10/967,668 entitled "Surgical Access System and Related Methods," the entire contents of which are each incorporated herein by reference as if set forth herein in their entireties. With reference to FIGS. **3**-**4**, a discussion of the lateral access instruments and methods is provided in brief detail. With the patient **20** positioned on their side, a surgical access system **26** is advanced through an incision **20**, into the retroperitoneal space **22**, and then through the psoas muscle **24** until the targeted spinal site (e.g. the disc space between V1 and V2) is reached. The access system **26** may include at least one tissue dilator, and preferably includes a sequential dilation system **28** with an initial dilator **30** and one or more additional dilators **32** of increasing diameter, and a tissue retractor assembly **34**. As will be appreciated, the initial dilator **30** is preferably advanced to the target site first, and then each of the additional dilators **32** of increasing diameter are advanced in turn over the previous dilator. A k-wire (not shown) may be advanced to the target site and docked in place (for example, by inserting the k-wire into the vertebral disc) prior to, in concurrence with, or after advancing the initial dilator **30** to the target site. With the sequential dilation system **28** positioned adjacent the target site (and optionally docked in place via the k-wire), the retractor assembly **34** is then advanced to the target site over the sequential dilation system **28**. According to the embodiment shown, the retractor assembly **34** includes retractor blades **36**, **38**, **40** and a body **42**. With the sequential dilation system **28** removed, the retractor blades **36**, **38**, and **40** are separated (FIG. **5**), providing the lateral access corridor through which instruments may be advanced to prepare the disc space and insert the implant **10**. According to one example, the posterior blade **36** may be fixed in position relative to the spine prior to opening the retractor blades. This may be accomplished, for example by attaching a shim to the blade **36** (e.g. via track **44** including dove tail grooves formed on the interior of blade **36**) and inserting the distal end of the shim into the disc space. In this manner, the posterior blade **36** will not move posteriorly (towards nerve tissue located in the posterior portion of the psoas muscle). Instead, the blades **38** and **40** will move away from the posterior blade **36** to expand the access corridor. Additionally, nerve monitoring (including determining nerve proximity and optionally directionality) is preferably performed as each component of the access system **26** is advanced through the psoas muscle, protecting the delicate nerve tissue running through the psoas, as described in the '949 patent and '668 application. Monitoring the proximity of nerves also allows the posterior blade **36** of the retractor assembly **34** to be positioned very posterior (all the way back to the exiting nerve roots), thus exposing a greater portion of the disc space than would otherwise be safely achievable. This in turn permits full removal of the disc and implantation of a wider implant and the wider implant makes utilization of the hyper-lordotic implant with a large lordotic angle (e.g. between 20-40 degrees) more practical.

With the lateral access corridor formed (as pictured in FIG. **5**) the target site may be prepped for insertion of the implant **10**. Preparation of the disc space may include performing an annulotomy, removal of disc material, and abrasion of the endplates and instruments such as annulotomy knives, pituitaries, curettes, disc cutters, endplate scrapers may be used. Additionally, as discussed above, it may generally be necessary to release the ALL in order to create enough flexibility between the adjacent vertebrae (e.g. V1 and V2) to receive the hyper-lordotic implant **10**. Unlike an anterior approach (where the great vessels and other tissue lying anterior to the disc space is retracted during the approach), when the target disc is approached laterally, the great vessels remain adjacent to the ALL along the anterior face of the spine. Thus, while cutting the ALL is generally simple and necessary during an anterior approach surgery, cutting the ALL during a lateral approach surgery has typically been unnecessary and can be difficult because of the need to avoid damaging the great vessels. Accordingly, FIGS. **5**-**14** set forth various example embodiments of ALL resecting instruments for safely releasing the ALL from a lateral approach.

FIGS. **5**-**8** illustrate one example ALL resector **50**. The ALL resector **50** includes a tissue retractor **52** and a sliding blade **68** and functions to both sever the ALL and protect surrounding tissue (e.g. great vessels, nerves etc. . . .) from unwanted damage while the ALL is being severed. The tissue retractor **52** includes a handle **54**, hollow shaft **56**, and head **58**. The head **58** is curved, preferably such that the inside surface **60** complements the curvature of the anterior aspects of the spinal target site. The head **58** may thus be positioned through the lateral access corridor to the spine and such that the curved interior surface **60** nestles around the curved anterior aspect of the spine. The outside surface **62** will form a barrier, protecting tissue along the anterior spine from inadvertent contact with the sliding blade when the ALL is cut. Furthermore, the tissue retractor **52** can be further manipulated move tissue and further expose the anterior aspect of the target site. The hollow shaft **56** includes a central lumen **64** with an opening adjacent the head **58** and another opening at the opposing end such that the sliding blade **54** may travel through the shaft **56**.

The sliding blade **68** includes a blade **70** that is secured to the distal end of an extender **72** by way of an attachment feature **74**. The attachment feature **74** as shown is similar to known attachment features used for attaching a blade at the end of a scalpel. It will be appreciated that any number of mechanisms may be used to attach blade **70** to extender **72**. Blade **70** may be disposable and extender **72** may be reusable. Alternatively, both blade **70** and extender **72** may be reusable or both may be disposable. The blade **70** includes a cutting edge **76** that, when advanced beyond the lumen **64** of shaft **56**, cuts through tissue or material situated adjacent the cutting edge **76**.

The proximal end of the extender **72** includes a grip **78** that a user may use to manipulate the position of the sliding blade **40** relative to the shaft **56** and head **58**. At least one stop feature **80** extends from the outer surface of the extender **72** which engages with a track **66** that extends along a portion of the elongated shaft **56**. The track **66** limits the longitudinal travel of the sliding blade **68** relative to the shaft **56** so that the sliding blade **68** remains slidably mated to the tissue retractor **52** without becoming unassembled and such that the blade **70** cannot extend beyond the protective head **58**. Additionally, the stop feature **80** restricts rotation of the sliding blade **68** relative to the tissue retractor **52**.

FIGS. **9-11** illustrate another example embodiment of an ALL resector **82**. The ALL resector **82** includes a handle **84** (for example, a T-handle) located at the proximal end of the elongated shaft **86** and a distal head **88** for resecting the ALL. The distal head **88** includes distally extending first and second fingers **90**, **92**, which form an opening **94** therebetween. First and second tapered surfaces **96**, **98** which extend a distance from the elongated shaft **84** along the fingers **90**, **92** enable the distal head **88** to insert gently between tissue. As best shown in FIG. **11**, the first finger **90** may be shorter in length than the second finger **92**. However, the first and second finger **90**, **92** may be provided in any number of length configurations without departing from the scope of the present invention. By way of example, it has been contemplated that the first finger **14** may be completely removed. Alternatively, as pictured in FIG. **12**, the fingers **90** and **92** may be curved and have a more substantial width than shown in FIGS. **9-11**. The curvature of the first and second fingers allows the distal head **88** to follow closely along the anterior side of the spine and/or along a curved spatula (not shown) positioned adjacent the anterior side of the vertebral body. Though not shown, a user may optionally insert a spatula along the anterior portion of the ALL prior to inserting the ALL retractor **82**. The spatula may serve as additional protection between the delicate tissue anterior to the ALL and the cutting blade **97** of the ALL resector **82**. With a spatula in place the user may insert the distal head **88** such that it approaches the lateral side of the ALL and is guided along the inside edge of the spatula. By way of example, the spatula may be straight or curved to match the selected fingers of the distal head **88**.

A cutting blade **97** is exposed between the first and second fingers **90**, **92** in the opening **17**. A slot **99** formed along a side of the distal head **88** allows a cutting blade **97** to be inserted and removed from the distal head **88**. Thus, the cutting blade **97** may be disposable and the remainder of the ALL resector **82** may be reusable. Alternatively, both cutting blade **97** and remainder of the ALL resector **82** may be reusable or both may be disposable. In use, the ALL resector **82** is preferably positioned such that the second finger **92** is aligned along the anterior side of the ALL and the first finger **90** is aligned along the posterior side of the ALL, thus, at least partially bounding the ALL on either side. The ALL resector **82** is advanced forward so that the cutting blade **97** cuts through the ALL from one lateral edge to the other. As discussed above, the second finger **92** is preferably aligned along the anterior side of the ALL as the distal head **88** is advanced, thereby shielding the tissue lying anterior to the finger **92** (e.g. great vessels, etc.) from the cutting blade **97**. Furthermore, as the user advances the ALL resector **82**, the fingers **90**, **92** may also act as a stabilizing guide.

Turning now to FIGS. **13-19**, various embodiments of the hyper-lordotic implant **10** are described. FIGS. **13-15**, for example, illustrate an example implant **110**. Implant **110** may preferably be comprised of any suitable non-bone composition having suitable radiolucent characteristics, including but not limited to polymer compositions (e.g. poly-ether-ether-ketone (PEEK) and/or poly-ether-ketone-ketone (PEKK)) or any combination of PEEK and PEKK. Other materials such as for example, metal, ceramics, and bone may also be utilized for the implant **110**. Implant **110** has a top surface **116** and bottom surface **118** for contacting V1 and V2, anterior sidewall **112**, posterior sidewall **114**, and front or leading side **120**, and rear or trailing side **122**. As discussed, the anterior sidewall **112** has a height greater than the posterior sidewall **114** such that the top surface **116** and bottom surface **118** converge towards each other in the posterior direction. By way of example, the top and bottom surfaces may converge at an angle between 20 and 40 degrees. It is contemplated that variations of the implant **110** may be simultaneously provided such that the user may select from different available ranges. For example, variations may be provided with 20 degree, 30 degree, and 40 degree angles. The top and bottom surfaces may be planar or provided as convex to better match the natural contours of the vertebral end plates. The length of the implant **110** is such that it may span from one lateral aspect of the disc space to the other, engaging the apophyseal ring on each side. By way of example, the implant **110** may be provided with a length between 40 mm and 60.

Having been deposited in the disc space, the implant **110** facilitates spinal fusion over time by maintaining the restored curvature as natural bone growth occurs through and/or past the implant **10**, resulting in the formation of a boney bridge extending between the adjacent vertebral bodies.

The implant **110** may be provided with any number of additional features for promoting fusion, such as apertures **127** extending between the top and bottom surfaces **116-117** which allow a boney bridge to form through the implant **110**. Various osteoinductive materials may be deposited within the apertures **127** and/or adjacent to the implant **110** to further facilitate fusion. Such osteoinductive materials may be introduced before, during, or after the insertion of the exemplary spinal fusion implant **110**, and may include (but are not necessarily limited to) autologous bone harvested from the patient receiving the spinal fusion implant, bone allograft, bone xenograft, any number of non-bone implants (e.g. ceramic, metallic, polymer), bone morphogenic protein, and bio-resorbable compositions, including but not limited to any of a variety of poly (D,L-lactide-co-glycolide) based polymers. Visualization apertures **128** situated along the sidewalls, may aid in visualization at the time of implantation and at subsequent clinical evaluations. More specifically, based on the generally radiolucent nature of the preferred embodiment of implant **110**, the visualization apertures **128** provide the ability to visualize the interior of the implant **110** during X-ray and/or other suitable imaging techniques. Further, the visualization apertures **128** will provide an avenue for cellular migration to the exterior of the implant **110**. Thus the implant **110** will serve as additional scaffolding for bone fusion on the exterior of the implant **110**.

The implant **110** also preferably includes anti-migration features designed to increase the friction between the implant **110** and the adjacent contacting surfaces of the vertebral bodies V1 and V2 so as to prohibit migration of the implant **110** after implantation. Such anti-migration features may include ridges provided along the top surface **116** and/or bottom surface **118**. Additional anti-migration features may also include spike elements **124** disposed along the top and bottom surfaces.

Tapered surfaces **126** may be provide along the leading end **120** to help facilitate insertion of the implant **110**. Additional instrumentation may also be used to help deploy the implant **110** into the disc space. By way of example, the implant installation device shown and described in detail in the commonly owned and co-pending U.S. patent application Ser. No. 12/378,685, entitled "Implant Installation Assembly and Related Methods," filed on Feb. 17, 2009, the entire contents of which is incorporated by reference herein, may be used to help distract the disc space and deposit the implant therein.

FIGS. **16-17** illustrate an implant **210** according to another example embodiment. The implant **210** share many similar features with the implant **110** such that like feature are numbered alike and repeat discussion in not necessary. The implant **210** differs from the implant **110** in that a trailing side **212** is configured for fixed engagement to one of the adjacent vertebral bodies (i.e. V1 or V2) to supplement the anti migration features and ensure the hyper-lordotic implant is not projected out of the disc space. Specifically, the implant **210** includes a tab **213** extending vertically above the top surface **116** and/or below the bottom surface **118**. Tabs **213** facilitate fixation of the implant **210** to the adjacent vertebrae. In the example shown in FIGS. **16-17**, the tab **213** extends past both the bottom and top surfaces **118** and **116**, respectively. It will be appreciated, however, that the implant **210** may include, without departing from the scope of the present invention and by way of example only, a tab **213** that extend solely in one direction. Additionally, multiple tabs **210** of various sizes and/or shapes may be used throughout a single implant **210** to facilitate implantation or fixation. In the illustrated example, the tab has a single fixation aperture **214** for enabling the engagement of a fixation anchor (not shown) within the vertebral bone to secure the placement of the implant. In use, when the implant **210** is positioned within the disc space, the tab **213** engages the exterior of the upper and lower vertebra and the fixation anchor may be driven into the side of either the upper or lower vertebra. One will appreciate that various locking mechanism may be utilized and positioned over or within the fixation aperture **214** to prevent the fixation anchor from unwanted disengagement with the implant **210**. For example, a suitable locking mechanism may be in the form of a canted coil disposed within the fixation aperture **214** or a screw that may be engaged to the trailing end **212** and cover all or a portion of the fixation aperture **214** after the fixation anchor positioned.

FIGS. **18-19** illustrate an implant **310** according to another example embodiment. The implant **310** shares many similar features with the implants **110** and **110** such that like feature are numbered alike and repeat discussion in not necessary. The implant **310** differs from the implant **110** in that a trailing side **312** is configured for fixed engagement to one of the adjacent vertebral bodies (i.e., V1 or V2) to supplement the anti-migration features and ensure the hyper-lordotic implant is not projected out of the disc space. Specifically, the implant **310** includes a tab **313** extending vertically above the top surface **116** and/or below the bottom surface **118**. Tab **313** facilitates fixation of the implant **310** to the adjacent vertebrae. Tab **313** extends past both the bottom and top surfaces **118** and **116**, respectively. It will be appreciated, however, multiple tabs **310** of various sizes and/or shapes may be used throughout a single implant **310** to facilitate implantation or fixation. The implant **310** differs from the implant **210** in that the tab **213** has includes a pair of fixation apertures **214** for enabling the engagement of a pair fixation anchors (not shown) within the vertebral bone to secure the placement of the implant. In use, when the implant **310** is positioned within the disc space, the tab **313** engages the exterior of the upper and lower vertebra and a fixation anchor is driven into the side of each of the upper or lower vertebra. One will appreciate that various locking mechanism may be utilized and positioned over or within the fixation apertures **314** to prevent the fixation anchors from unwanted disengagement with the implant **310**. One will also appreciate that single locking mechanism may be used to cover both fixation apertures **314**, or, a separate locking mechanism may be used to cover each fixation aperture **314**. Again, by way of example, a suitable locking mechanism may be in the form of a canted coil disposed within each fixation aperture **314** or a screw that may be engaged to the trailing end **312** and cover all or a portion of each fixation aperture **314** after the fixation anchors are positioned.

While this invention has been described in terms of a best mode for achieving this invention's objectives, it will be appreciated by those skilled in the art that variations may be accomplished in view of these teachings without deviating from the spirit or scope of the invention. For example, particularly at L5-S1 where the pelvic bone makes a lateral access approach difficult, an antero-lateral approach similar to the approach utilized during appendectomies may be utilized.

What is claimed is:

1. A resector instrument comprising:

a tissue retractor comprising:

- a handle,
- a hollow shaft coupled to the handle, the hollow shaft having a central lumen therein and defining a longitudinal axis,
- a first opening at a proximal end of the central lumen,
- a second opening at a distal end of the central lumen,
- a longitudinal track disposed on the hollow shaft, and
- a curved head adjacent the second opening; and

a sliding blade disposed within the central lumen and configured to move axially therein,

wherein the sliding blade includes a stop slidably coupled to the longitudinal track and configured to prevent rotation of the sliding blade relative to the hollow shaft,

wherein the sliding blade is configured to sever tissue in an extended position, and the curved head is configured to protect surrounding tissue from contact with the sliding blade.

2. The resector instrument of claim **1**, wherein the tissue is a ligament.

3. The resector instrument of claim **2**, wherein the ligament is the Anterior Longitudinal Ligament (ALL).

4. The resector instrument of claim **2**, wherein the extender further comprises a grip disposed at a proximal end thereof, the grip being configured to control an axial position of the sliding blade relative to the shaft and the curved head.

5. The resector instrument of claim **2**, wherein rotation of the sliding blade is restricted relative to the tissue retractor.

6. The resector instrument of claim **5**, further comprising:

a track disposed on the hollow shaft and extending along a portion thereof; and

a stop feature disposed on the extender and extending from an outer surface thereof,

wherein the stop feature is configured to engage with the track, and

wherein the track is configured to limit axial movement of the sliding blade relative to the hollow shaft.

7. The resector instrument of claim **1**, wherein the sliding blade comprises:

an extender; and

a blade secured to a distal end of the extender,

wherein the blade includes a distal cutting edge that, when advanced distally beyond the second opening, is configured to sever the tissue adjacent the cutting edge.

8. The resector instrument of claim **7**, wherein the blade is configured to be disposable.

9. The resector instrument of claim **7**, wherein the extender is configured to be reusable.

10. The resector instrument of claim **7**, wherein the blade and the extender are configured to be reusable.

11. The resector instrument of claim **7**, wherein the blade and the extender are configured to be disposable.

12. The resector instrument of claim **1**, wherein an inner surface of the curved head is configured to complement a curvature of an aspect of a target site.

13. The resector instrument of claim **12**, wherein the inner surface of the curved head is configured to complement a curved anterior aspect of a spine of a patient.

14. The resector instrument of claim **1**, wherein an outer surface of the curved head is configured to form a barrier between the sliding blade and a non-target tissue, thereby protecting the non-target tissue from inadvertent contact with the sliding blade.

15. The resector instrument of claim **14**, wherein the non-target tissue is tissue along an anterior aspect of a spine of a patient.

16. The resector instrument of claim **1**, wherein the tissue retractor is configured to manipulate tissue and further expose an aspect of a target site.

17. The resector instrument of claim **1**, wherein the longitudinal track is configured to prevent the sliding blade from extending past a distal end of the curved head.

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